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INTRODUCTION :



- Amperometry is one of the electrochemical methods.
- It is concerned with the measurement of current under constant applied voltage.
- It is a form of quantitative analysis.
- When indicator method is not suitable we use amperometric method for determination of end point.
- Amperometric titration are otherwise called as polarographic titration because of similarity in principle.



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PRINCIPLE:



- In amperometric titration, the potential applied between the indicator electrode and the appropriate depolarising reference electrode is kept constant & current pass through the cell, it is then measured during the titration.



- Indicator electrode : Dropping mercury electrode. (DME)
- Reference electrode : Saturated calomel electrode

- During the titration the concentration of electro-reducible ion changes & hence the diffusion current also changes.



- According to Ilkovic equation :



$$I_d = 607 n C D^{1/2} m^{2/3} t^{1/6}$$

where,

I_d = Diffusion current due to electro-reducible ions.

n = No. of electrons involved in the reduction of one molecule .

C = Concentration expressed in mmol/ lit .

D = Diffusion coeff. of ions (cm²/sec).



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$m =$ Wt. of mercury flowing through capillary
 (mg/sec)
 $t =$ Drop time in second.



CONDITION FOR PERFORMING AMPEROMETRIC TITRATION:

- Both should be reducible .
- the potential applied should limiting current.

APPARATUS USED FOR AMPEROMETRIC TITRATION :

- Dropping mercury electrode :
- Capillary tube about 10 - 15 cm.
- Internal diameter of capillary 0.05mm.
- A vertical distance being maintained between DME & the solution.
- Drop time : 1-5 se
- Drop diameter – 0.5 mm

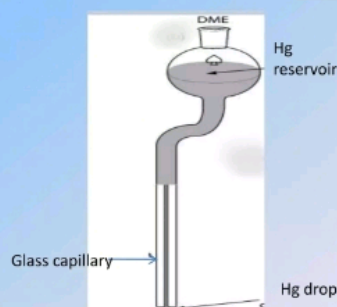


Fig. Dropping Mercury Electrode



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• Rotating Platinum Micro-electrode :

- It consists of a glass rod with a bent platinum wire at about 600 rpm .
- Wire contacts are made through a mercury reservoir at the top so the potential can be applied & the current is produced

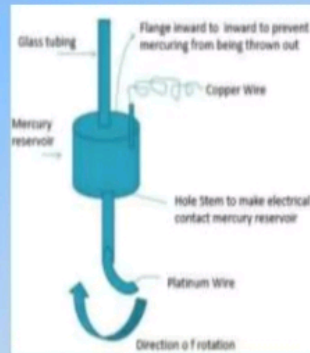


Fig. Rotating Platinum Micro-Electrode

Advantages of Using RPME over DME :

- Mercury cannot be used as electrode at positive potentials because of its oxidation, RPME is used.
- Diffusion current is 20 times larger than DME which allows to measure the small concentration of ion.
- The rotating platinum electrode can be used at positive potential up to + 0.9 Volt whereas DME can be used only +0.4 volt to -2.0 Volt.
- The electrode is simple to construct.

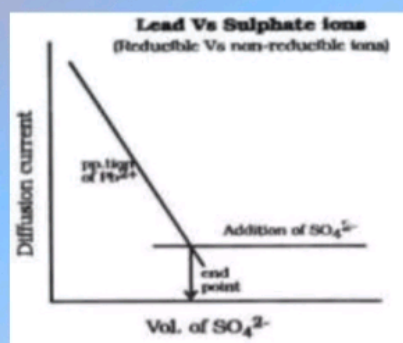


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Types of amperometric titrations:

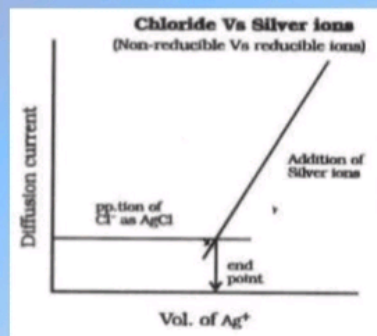
- Titration of reducible ions vs non reducible ions.
eg . Lead (Pb^{2+}) vs sulphate ions (SO_4^{2-})
- Titration of non reducible ions vs reducible ions.
eg . Chloride (Cl^-) vs silver (Ag^+)
- Titration of reducible ion vs reducible ions.
eg. Lead (Pb^{2+}) vs Dichromate ion ($\text{Cr}_2\text{O}_7^{2-}$)
- Redox titration (oxidant and reductant).
eg. Ferric (Fe^{3+}) ions vs titanous ions (Ti)
- End point techniques (Karl Fischer).
(Determination of water using Karl Fischer reagent)

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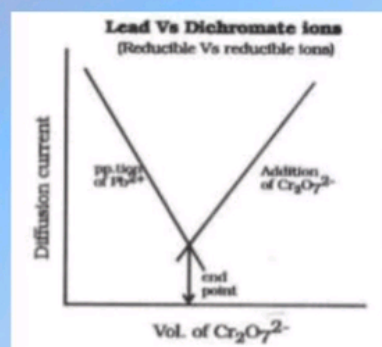


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- Titration of non reducible ions vs reducible ions.
eg . Chloride (Cl^-) vs silver (Ag^{2+})

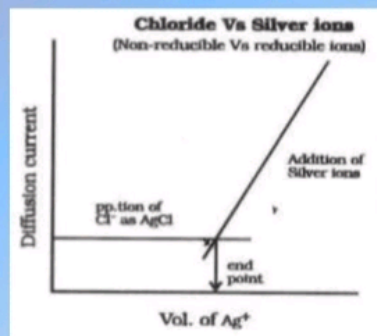


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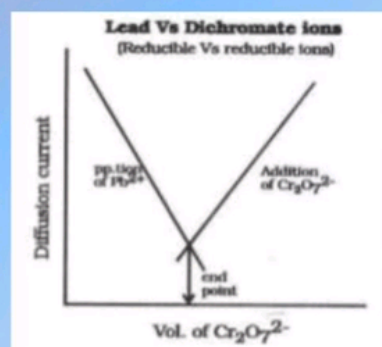


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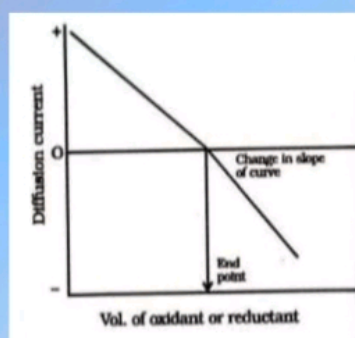
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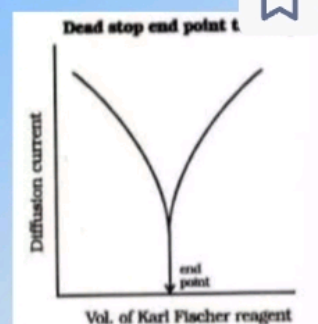
- Redox titration (oxidant and reductant).
eg. Ferric (Fe^{3+}) ions vs titanous ions (Ti)



- End point techniques (Karl Fischer).
(Determination of water using Karl Fischer reagent) :



- A small potential is applied between the two similar platinum electrodes.
- Addition of KF reagent (solution of iodine and SO_2 in pyridine and methanol) then till the end point. Where current is decrease at end point and only one electrode is depolarised and current is zero.



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Advantages of Amperometric titration

- Both reducible as well as non- reducible ion groups can be determine. 
- Dilute solution can be analysed.
- The reaction carryout can be reducible or irreversible.
- The appratus is simple and temprerature need not be provided constants.

Application of Amperometric titration

- Amperometric titration : 
 - quantitative in nature.
 - used and determine the end point.
- Determination of water by using karl fischer reagent.
- Amperometric dectector - (HPLC)Amperometric dectector can detect very low conc. of reducible ions & they can easily determine.
- Quantification of ion or mixture of ions.



Disadvantages of amperometric titration



- Inaccurate result are sometime obtained.
- The foreign substance should not be present in a larger concentrations than the substance to be titrated.

Reference :



- Textbook of pharmaceutical analysis , 4th edition by Dr. Ravi Sankar.
- Instrumental method of chemical analysis by G. R. Chatwal & Sham K. Anand , hmalaya publishing house.



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